

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103
Semester Examination – May - 2019

Branch: Mechanical Engineering

Sem.:- IV

Subject with Subject Code:- Numerical Method in Mechanical Engineering (BTMEC404)

Marks: 60

Date:- 22/5/2019

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

Q.1. a) Round off the following to four significant digits

04

3.26425, 35.46735, 0.00032217, 6.248359

b) The measuring length of bridge and rivet are 9999 and 9 cm respectively, whereas the true value are 10,000 and 10 cm respectively. Compute true error and true percent relative error for each case.

04

c) A body travels uniformly a distance of (13.8 ± 0.2) in a time (4.0 ± 0.3) find velocity of body with in error limits.

04

Q.2. a) Explain the concept of bisection method graphically and write the necessary conditions to find out the real root by bisection method.

06

b) The volume V of liquid in a spherical tank of radius r is related to the depth h of the liquid by

$$V = \frac{\pi h^2(3r-h)}{3}. \text{ Determine } h \text{ using Newton-Raphson Method, given } r=1 \text{ m, and } V=0.5 \text{ m}^3$$

06

Q.3. a) Use Cramer's rule to solve system of equation

04

$$-x + 3y - 2z = 5; \quad 4x - y - 3z = -8; \quad 2x + 2y - 5z = 7.$$

b) Solve the system of equation with pivoting with pivoting

04

$$0.0003120x + 0.006032y = 0.003328; \quad 0.5000x + 0.894y = 0.9471$$

undefined

c) Monthly Faculty salary in three departments of an institute is given below. Assuming that salary for particular category is same in all the departments calculate the salary of each category of faculty 04

Department	Number of Faculty			Total Salary(thousands)
	Professor	Associate professor	Assistant Professor	
A	2	2	4	60
B	3	1	2	50
C	1	4	3	60

Q.4.) a) A solid of revolution is formed by rotating about x-axis the area between the x-axis, the line $x=0$ and $x=1$ and curve through the points with the following co-ordinates: 06

X	0.00	0.25	0.50	0.75	1.00
Y	1.000	0.9896	0.9589	0.9089	0.8415

Estimate the volume of the solid formed, giving answer to three decimal place by using Trapezoidal rule.

b) Velocity of car running on straight road at interval of 2 minutes are given below. 06

Time (T) (min)	0	2	4	6	8	10	12
Velocity(V) (km/hr)	0	22	30	27	18	7	0

Find the distance covered by car using Simpson's 1/3 rule

Q.5a) Given the equation $\frac{dy}{dx} = \frac{2y}{x}$ with $y(1) = 2$, estimate $y(2)$ using Euler's method using $h=0.25$. 04

b) The experimental values relating centripetal force and radius, for a mass travelling at constant velocity in a circle, are as shown:

Force(N)	5	10	15	20	25	30	35	40
Radius (cm)	55	30	16	12	11	9	7	5

Determine the equations of the regression line of force on radius. 04

c) Given that $\frac{dy}{dx} = y - x$ where $y(0) = 2$, find $y(0.1)$ with $h=0.1$ correct to four decimal place by using Range-Kutta fourth order method 04

Q.6. a) Write the algorithm for finding the real root single root of an equation by Bisection Method 04

b) Write the algorithm for to find the integration using Simpson's 1/3 rule 04

c) Write the algorithm for to solve ordinary differential equation using Euler's Method. 04

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL
UNIVERSITY, LONERE - RAIGAD -402 103
Winter Semester Examination - Dec - 2019**

Branch: Mechanical Engineering

Subject:- Numerical Methods in Mechanical Engineering (BTMEC404)

Date:- 02/12/2019

Sem.:- IV

Marks: 60

Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

(Marks)

- Q.1. (a)** Explain the following; Approximate error, Precision and accuracy with suitable example? (6)
- (b) (i)** The length and breadth of a rectangle are A body travels uniformly a distance of (5.7 ± 0.1) cm and (3.4 ± 0.2) cm. Find the area of the rectangle with in error limit. (4)
- (ii)** Round off the following to four significant digits. (2)
 $0.0063945, 0.090038$
- Q.2.** Find the positive root of the equation $xe^x = 1$ using bisection method which lies between 0 and 1 till approximate error becomes 10%. Show lower limit, upper limit and approximate error in each iteration. (12)
- Q.3.** Use Gauss elimination to solve (12)
$$\begin{aligned} 3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\ 0.1x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.3x_1 - 0.2x_2 + 10x_3 &= 71.4 \end{aligned}$$
- Q.4. (a)** The work done on an object is equal to the force times the distance moved in the direction of the force. The velocity of an object in the direction of a force is given by
$$v = 4t \quad 0 \leq t \leq 6$$

where v is in m/s. Employ the multiple application trapezoidal rule to determine the work if a constant force of 200 N is applied for all t . (8)
- (b)** Obtain the forward Difference expression for first derivatives. (4)
- Q.5. (a)** Solve the following ODE using Euler's method from $t = 0$ to 3 taking step size of 1.
$$\frac{dy}{dt} = -y + t \quad \text{given } y(0) = 1$$
 (6)

- (b) The table below gives the temperature T ($^{\circ}\text{C}$) and length l (mm) of a heated rod. Find the values of length at 55°C using quadric interpolation. (6)

T	40	50	60
l	600.5	600.6	600.8

- Q.6.** Draw the algorithm of the following Numerical Methods (Any Two). (12)
- (i) NR Method
 - (ii) Simple Trapezoidal Rule
 - (iii) $1/3^{\text{rd}}$ Simpson Rule

Paper End

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech.

Branch : Mechanical

Semester : IV

Subject Code & Name: BTMPE405A- Numerical Methods in Engg.

Max Marks:60

Date: 27/8/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All Questions are Compulsory
2. Draw neat diagram wherever necessary.
3. Figures to right indicates full marks
4. Assume suitable data wherever necessary and mention it clearly

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		
A) Define and Explain the precision and Accuracy with an example	CO1	06
B) The discharge Q over a notch for head H is calculated by the formula $Q = kH^{5/2}$ where 'k' is given constant. If the head is 75cm and an error of 0.15cm is possible in its measurement, estimate the percentage error in computing the discharge.	CO1	06
C) Explain with an example error Propagation with example	CO1	06
Q.2 Solve Any Two of the following.		
A) The bacteria concentration (C) in a reservoir varies as $C=4e^{-2t}+e^{-0.1t}$. Calculate the time required for bacteria concentration to be 0.5	CO2	06
B) Using Bisection method find positive root for $\sqrt{32}$ upto three decimal places	CO3	06
C) Using Newton Raphson method, find a root upto three decimal place for the $x^3+2x^2+10x-20=0$ take $x_0=1$	CO2	
Q.3 Solve Any Two of the following.		
A) Solve by Gauss Elimination Method	CO2	06
$\begin{aligned}10x - 7y + 3z + 5u &= 6 \\-6x + 8y - z - 4u &= 5 \\3x + y + 4z + 11u &= 2 \\5x - 9y - 2z + 4u &= 7\end{aligned}$		
B) Solve by Crammers rule	CO2	06
$\begin{aligned}x + 3y + 6z &= 2 \\3x - y + 4z &= 9 \\x - 4y + 2z &= 7\end{aligned}$		
C) A fruit seller selling different size mangoes. He sell a small size magno for Rs 3/-, medium size mango for Rs 5/-, and large size mango for Rs 7/-. The seller sells total 15 mangos and has made Rs 77/-. He has sold 2 more medium size mangos than small mangos. How many of each size did he sell:	CO3	06

Q.4 Solve Any Two of the following.

CO2 06

A) Evaluate the $\int_0^1 \left(\frac{1}{x^2+1} \right) dx$ by using two point Simpsons 1/3 rule

B) Evaluate the $\int_0^{\pi/2} \left(\frac{x}{\sin x} \right) dx$ using Trapezoidal method

CO2 06

C) A train is moving at the speed of 30 m/s. Suddenly the brakes are applied. The speed of the train per second after 't' is given by

CO3 06

Time 't' in second	0	5	10	15	20	25	30	35	40	45
Speed 'v' in m/s	30	24	19	16	13	11	10	8	7	5

Estimate the distance moved by the train in 45 seconds

Q.5 Solve Any Two of the following.

CO3 06

A) A chemical factory wishing to study the effect of the extraction time (T) on the efficiency of an extraction operation (E) obtained data is as shown in the table. Fit a straight line to given data by least square method

T	27	45	41	19	35	39	19	49	15	31
E	57	64	80	46	62	72	52	77	57	68

B) Given $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y(0)=1$. Find $y(0.1)$ using modified Eulers method

CO2 06

C) Find $y(0.2)$ by using Runge-Kutta method taking $h=0.1$ and $y(0)=1$ from the system of equation, $dy/dx = x + y^2$

CO2 06

*** End ***

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Write a note on	CO1	6
i) True error ii) True percent relative error		
iii) Approximate percent relative error		
B) The exponential function can be computed using	CO1	6
$e^x = 1 + x + x^2/2! + x^3/3! + \dots + x^n/n!$		
add terms one at a time to estimate $e^{(0.2)}$ After each new term is added, compute the true and approximate percent relative errors and make a table. Add terms until the absolute value of the approximate error estimate 'ea' falls below a pre specified error criterion 'es' conforming to three significant figures.		
C) Write a note on	CO1	6
i) Truncation error and Taylor series ii) Machine epsilon, iii) Accuracy and Precision		
Q.2 Solve Any Two of the following.		12
A) The velocity v of a falling parachutist is given by	CO2	6
$v = gm/c(1 - e^{-(c/m)t})$		
where $g = 9.8 \text{ m/s}^2$. For a parachutist with a drag coefficient $c = 15 \text{ kg/s}$, compute the mass 'm' so that the velocity is $v = 35 \text{ m/s}$ at $t = 9 \text{ s}$. Determine 'm' to a level of $\epsilon_s = 0.1\%$. Using Newton Raphson		
B) You are designing a spherical tank to hold water for a small village in a developing country. The volume of liquid it can hold can be computed as	CO2	6
$V = \pi h^2(3R - h)/3$, where $V = \text{volume [m}^3\text{]}$, $h = \text{depth of water in tank [m]}$,		

and R = the tank radius [m]. If $R = 3$ m, to what depth must the tank be filled so that it holds 30 m^3 Use Bisection method

- C) Write a note on i) Stopping criteria ii) Error propagation iii) Significant digits 6

Q. 3 Solve Any Two of the following. 12

- A) Given the system of equations CO3 6

$$-3x_2 + 7x_3 = 2$$

$$x_1 + 2x_2 - x_3 = 3$$

$$5x_1 - 2x_2 = 2$$

(a) Compute the determinant.

(b) Use Cramer's rule to solve for the x 's.

- B) For the following set of simultaneous equations CO3 6

$$100a + T = 519.72$$

$$50a - T + R = 216.55$$

$$25a - R = 108.27$$

Solve for acceleration a and the tensions T and R

- C) The following system of equations was generated by applying the mesh current law to the circuit CO3 6

$$60I_1 - 40I_2 = 200$$

$$-40I_1 + 150I_2 - 100I_3 = 0$$

$$-100I_2 + 130I_3 = 230$$

Solve for I_1 , I_2 , and I_3 .

Q.4 Solve Any Two of the following. 12

- A) Evaluate the following integral: CO4 6

$$\int_0^{\pi/2} (8 + 4 \cos x) dx$$

(a) analytically; (b) single application of the trapezoidal rule;

(c) multiple-application trapezoidal rule, with $n = 2$ and 4 ; For each of the numerical estimates (b) through (c), determine the percent relative error based on (a)

- B) Suppose that the upward force of air resistance on a falling object is proportional to the square of the velocity. For this case, the velocity can be computed as CO4 6

$$v(t) = \sqrt{\frac{gm}{cd}} \tanh\left(\sqrt{\frac{gcd}{m}} t\right)$$

where c_d = a second-order drag coefficient. (a) If $g = 9.8 \text{ m/s}^2$, $m = 68.1 \text{ kg}$ and $c_d = 0.25 \text{ kg/m}$, use analytical integration to determine how far the object falls in 10 s. (b) Make the same evaluation, but evaluate the integral with the multiple-segment trapezoidal rule. Use a sufficiently high n that you get three significant digits of accuracy

- C) The following data was collected for the distance traveled

CO4

6

versus time for a rocket:

t, s	0	25	50	75	100	125
y, km	0	32	58	78	92	100

Use numerical differentiation to estimate the rocket's velocity and acceleration at each time

- Q. 5 Solve Any Two of the following.

12

- A) An object is suspended in a wind tunnel and the force measured for various levels of wind velocity. The results are tabulated below.

CO5

6

v, m/s	10	20	30	40	50	60	70	80
F, N	25	70	380	550	610	1220	830	1450

Use least-squares regression to fit this data with (a) a straight line, (b) a power equation based on log transformations, and (c) a power model based on nonlinear regression. Display the results graphically

- B) Use least-squares regression to fit a straight line to

CO5

6

x	1	2	3	4	5	6	7	8	9
y	1	1.5	2	3	4	5	8	10	13

- C) Explain the basic structure of a C program with suitable example.

CO5

6

*** End ***

Course: B. Tech.

Branch : MECHANICAL ENGINEERING & ALLIED

Semester : IV

Subject Code & Name: BTMPE405A- Numerical Methods in Engineering

Max Marks: 60

Date:25-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

12

- A) Explain the concepts of precision and accuracy in the context of numerical analysis. How do significant figures contribute to representing precision in measured values? Provide an example to illustrate the difference between a precise and accurate measurement. CO1 6
- B) Discuss the concept of truncation error in numerical methods and its relationship with Taylor series approximation. Explain how the accuracy of an approximation is affected by truncation error. Provide a mathematical example to illustrate the concept. CO1 6
- C) Use zero- through fourth-order Taylor series expansions to approximate the function

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$
 From $x_i = 0$ with $h = 1$. That is, predict the function's value at $x_{i+1} = 1$. CO1 6

Q.2 Solve Any Two of the following.

12

- A) Estimate the root of the equation:

$$f(x) = e^{-x} - x$$
 Using the **Newton-Raphson method**, employing an initial guess of $x_0 = 0$. CO2 6
- B) Calculate the roots of the following equation, using **Bi-Section Method**.

$$f(x) = x^3 - 5x + 1$$
, given that the roots lies between 0 and 1. CO2 6
- C) Estimate the approximate value of $17^{(1/3)}$ using **Newton-Raphson Method** with the initial approximation of $X_0 = 2$ CO2 6

Q. 3 Solve Any Two of the following.**12**

- A) Use Gauss elimination to solve
- $$0.0003x_1 + 3.0000x_2 = 2.0001$$
- $$1.0000x_1 + 1.0000x_2 = 1.0000$$
- B) Solve the system of the linear algebraic equations using:

CO3 **6****Cramer's rule**

$$0.3x_1 + 0.52x_2 + x_3 = -0.01$$

$$0.5x_1 + x_2 + 1.9x_3 = 0.67$$

CO3 **6**

$$0.1x_1 + 0.3x_2 + 0.5x_3 = -0.44$$

- C) Solve the system of the linear algebraic equations by using:

Naive Gauss Elimination method

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$

CO3 **6**

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$

Q.4 Solve Any Two of the following.**12**

- A) Evaluate the following integral by using composite trapezoidal rule with 2 subintervals and 4 subintervals

$$\int_0^1 \left(\frac{1}{1+x} \right) dx$$

CO3 **6**

- B) Numerically integrate the following equation from a = 0 and b = 0.8
By using

1. Simpson's 1/3 Rule andCO3 **6****2. Simpson's 3/8 Rule**

$$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$$

- C) Estimate the first order derivative of the following polynomial using

- 1) Forward difference approximation
- 2) Backward difference approximation
- 3) Central difference approximation

CO3 **6**

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.25$$

Q. 5 Solve Any Two of the following.

12

- A) Use the method of **least squares** to fit the curve $Y = C_0/X + C_1\sqrt{X}$ to the table of values given below and calculate the values of C_0 and C_1

X	0.1	0.2	0.4	0.5	1	2
Y	21	11	7	6	5	6

CO4

6

Using the Second Order **Runge- Kutta Method**, numerically integrate the following polynomial by using the following

- 1.) Heun Method.
- 2.) The Midpoint Method.
- 3.) Ralston's Method

CO4

6

B)

$$f(x, y) = -2x^3 + 12x^2 - 20x + 8.5$$

For $x=0$ to $x=0.5$, with step size of 0.5.

The initial condition is $x=0$ at $y = 1$.

Using “**Polynomial Regression Methods**” formulate the equations of a second order polynomial and fit it for the following data points

C)

X	0	1	2	3	4	5
Y	2.1	7.7	13.6	27.2	40.9	61.1

CO4

6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,

LONERE-RAIGAD-402103

Regular and Supplementary Examination Summer -2024

Course: B. Tech

Semester: IV

Branch: Mechanical and Allied/ Production/ Automobile/ Automation and Robotics / Robotics / Mechatronics

Subject Name with Code: Numerical Methods in Engineering BTMPE405A

Max Marks: 60

Date: 24/06/2024

Duration: 3 hrs

Instructions to the Students

1. All the Questions are **compulsory**.
2. Use of non –a programmable calculator is **allowed**
3. Illustrate your answers with neat sketches, diagrams etc., wherever necessary
4. **Assume suitable data** with justification, in case the data provided is insufficient to solve problem

Q. No.		Level/CO	Marks
Q.1	Solve any two of the following		
A)	Define, True percentage error, Approximate percentage error and rounding off with suitable examples.	Understand/ CO1	6
B)	(i) The length and breadth of a rectangle are (5.7 ± 0.1) cm and (3.4 ± 0.2) cm. Find the area of the rectangle with in error limit. (ii) Round off the following to four significant digits. 3.63042, 0.063945, 0.0090038	Apply & Understand/ CO1	3+3
C)	(i) If $X = 0.51$ and is corrected to 2 decimal places. Find the relative error. (ii) Evaluate $\frac{25.2 \times 1374}{33.3}$ taking into account the significant digits.	Understand/ CO1	3+3
Q.2	Solve any two of the following		
A)	Clearly differentiate between Newton Raphson method and bisection method with suitable illustrations.	Understand/ CO2	6
B)	Find a real root of the equation $x = e^{-x}$, using Newton Raphson method with the initial guess as $x_0 = 0$ till the approximate error 20%.	Understand/ CO2	6
C)	Determine the real root of $\ln x^2 = 0.7$ using three iteration of the bisection method with the initial guesses of $x_l = 0.5$ and $x_u = 2$.	Apply/CO3	6
Q.3	Solve any two of the following		

A)	Given the system $\begin{aligned}x_1 + x_2 - x_3 &= -3 \\ 6x_1 + 2x_2 + 2x_3 &= 2 \\ -3x_1 + 4x_2 + x_3 &= 1\end{aligned}$ Solve by naïve Gauss Jordan with partial pivoting. Show all the steps of computation.	Understand/ CO2	6												
B)	Explain the concept of naïve Gauss elimination method taking three simultaneous linear equations.	Understand/ CO2	6												
C)	Explain the following terms WRT simultaneous linear equation: Ill condition, scaling and partial pivoting	Understand/ CO2	6												
Q.4	Solve any two of the following														
A)	Explain the $1/3^{\text{rd}}$ Simpson's numerical Integration method with suitable illustration and write its expression.	Understand/ CO2	6												
B)	The velocity of a car (running on straight road) at interval of 2 minutes are given below. <table border="1"><tr><td>Time (min)</td><td>0</td><td>2</td><td>4</td><td>6</td></tr><tr><td>Velocity (km/hr)</td><td>0</td><td>22</td><td>30</td><td>27</td></tr></table> Apply Simpson's $3/8$ rule and find distance covered by the car.	Time (min)	0	2	4	6	Velocity (km/hr)	0	22	30	27	Apply/ CO3	6		
Time (min)	0	2	4	6											
Velocity (km/hr)	0	22	30	27											
C)	Obtain the expression of the first order central numerical differentiation expression.	Apply/ CO3	6												
Q.5	Solve any two of the following														
A)	Use Euler's method to numerically integrate $\frac{dy}{dx} = -2x^3 + 12x^2 - 20x + 8.5$ using $h = 0.5$ with $y(0) = 1$ from $x = 0$ to 2	Understand/ CO3	6												
B)	i. Explain fundamental difference between least square regression and Interpolation. ii. Write the algorithm of bisection method to find the real root.	Remember/understand CO2/CO4	3+3												
C)	The table below gives the temperature T ($^{\circ}\text{C}$) and length l (mm) of a heated rod. If $l = a_0 + a_1T$, find the values of a_0 and a_1 using linear least squares <table border="1"><tr><td>T</td><td>40</td><td>50</td><td>60</td><td>70</td><td>80</td></tr><tr><td>l</td><td>600.5</td><td>600.6</td><td>600.8</td><td>600.9</td><td>601.0</td></tr></table>	T	40	50	60	70	80	l	600.5	600.6	600.8	600.9	601.0	Apply/ CO3	6
T	40	50	60	70	80										
l	600.5	600.6	600.8	600.9	601.0										

End